

# Introduction to Data Shaping and Grouping

INTRODUCTION TO POWER QUERY IN EXCEL

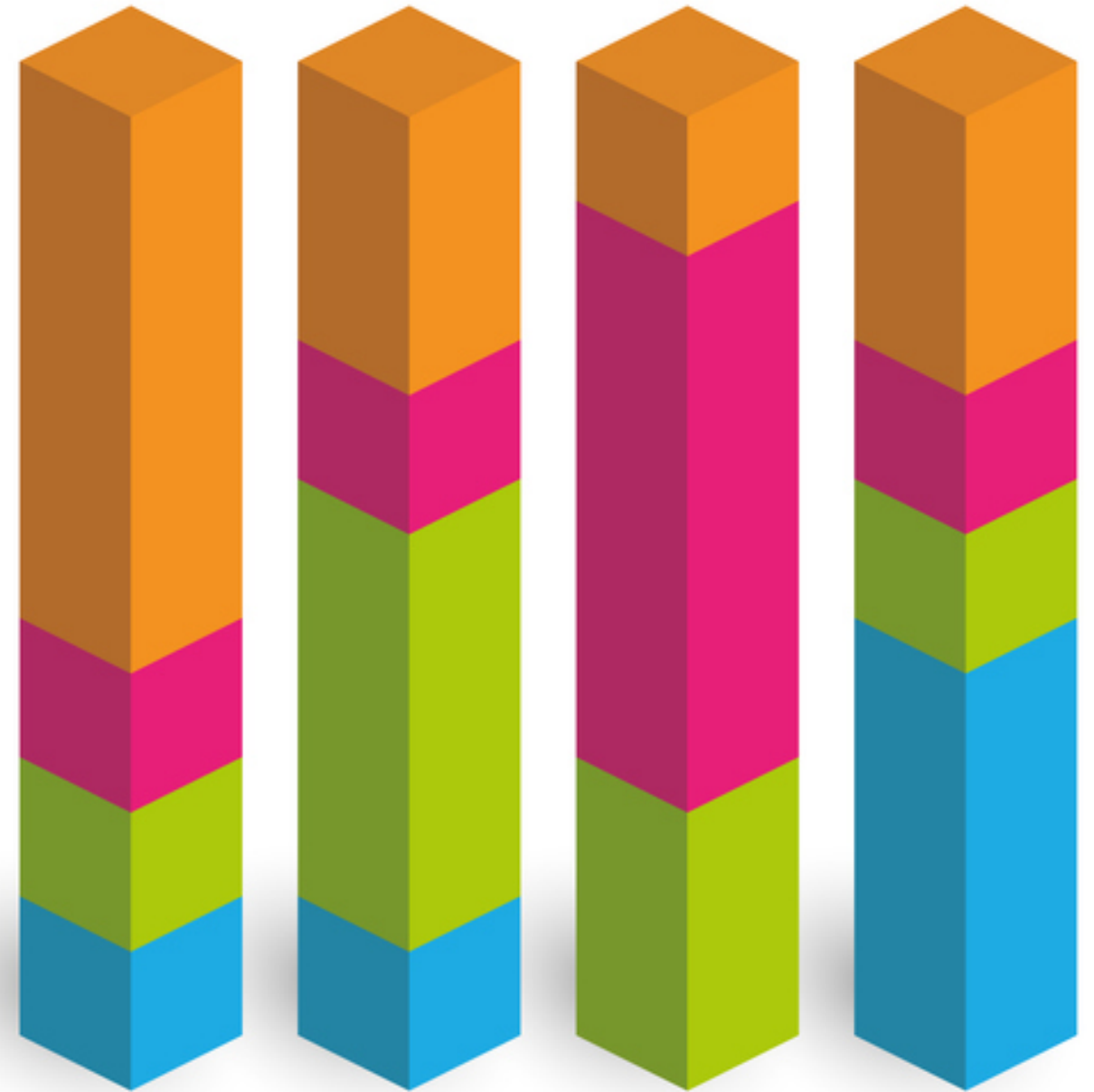


**Lyndsay Girard**

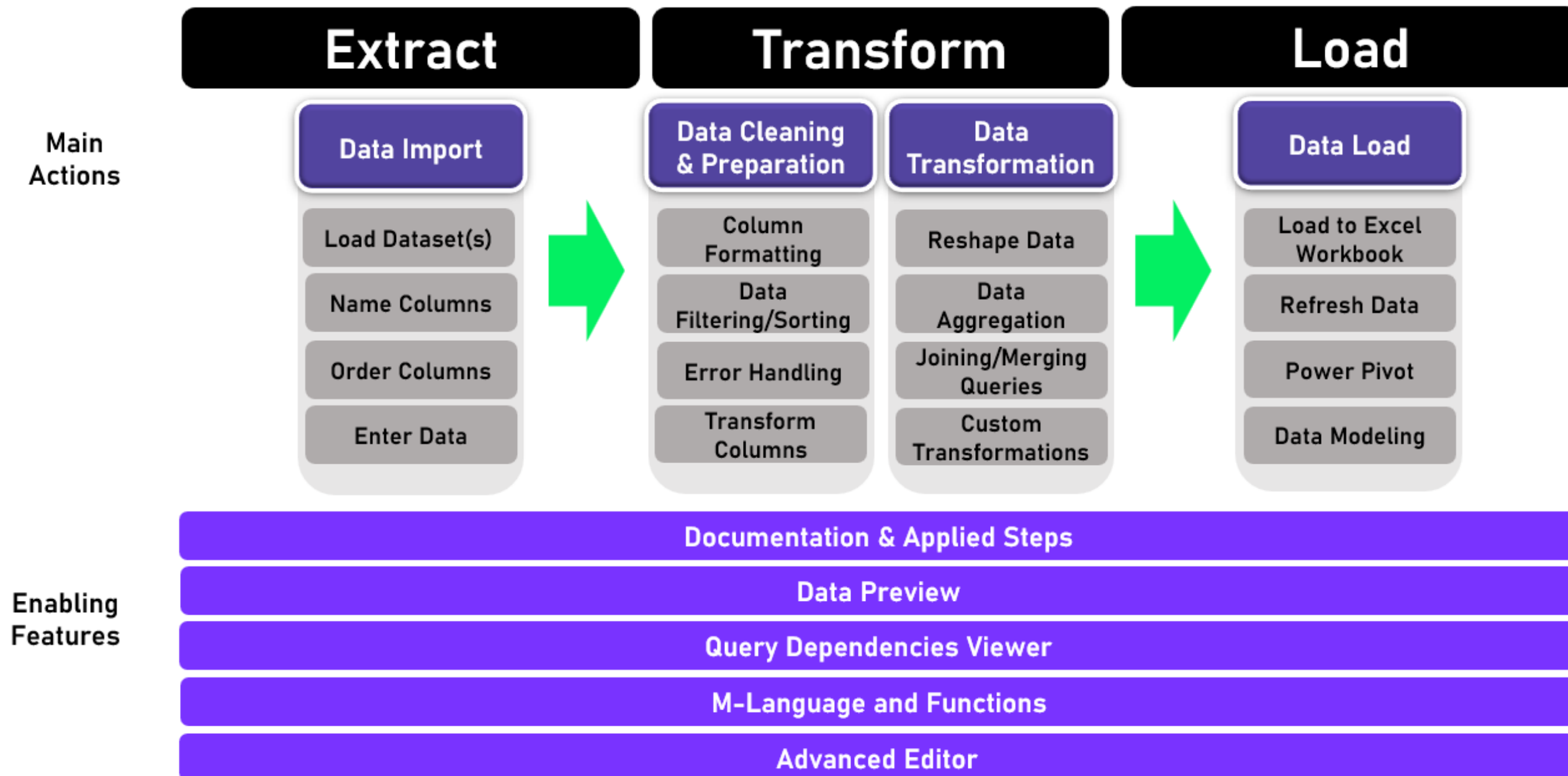
Performance Analytics Consultant

# Shaping and Grouping Data

- **Shaping:** restructuring data to make it more suitable for analysis.
- **Grouping:** Aggregating and summarizing data based on specific columns.



# Excel Power Query



# Excel Power Query

Main  
Actions

Extract

Data Import

Load Dataset(s)

Name Columns

Order Columns

Enter Data

Transform

Data Cleaning  
& Preparation

Column  
Formatting

Data  
Filtering/Sorting

Error Handling

Transform  
Columns

Data  
Transformation

Reshape Data

Data  
Aggregation

Joining/Merging  
Queries

Custom  
Transformations

Load

Data Load

Load to Excel  
Workbook

Refresh Data

Power Pivot

Data Modeling

Enabling  
Features

Documentation & Applied Steps

Data Preview

Query Dependencies Viewer

M-Language and Functions

Advanced Editor

# Excel Power Query

Main  
Actions

**Extract**

**Data Import**

Load Dataset(s)

Name Columns

Order Columns

Enter Data

**Transform**

**Data Cleaning  
& Preparation**

Column  
Formatting

Data  
Filtering/Sorting

Error Handling

Transform  
Columns

**Data  
Transformation**

Reshape Data

Data  
Aggregation

Joining/Merging  
Queries

Custom  
Transformations

**Load**

**Data Load**

Load to Excel  
Workbook

Refresh Data

Power Pivot

Data Modeling

Enabling  
Features

Documentation & Applied Steps

Data Preview

Query Dependencies Viewer

M-Language and Functions

Advanced Editor

# Excel Power Query

Main  
Actions

## Extract

### Data Import

Load Dataset(s)

Name Columns

Order Columns

Enter Data



## Transform

### Data Cleaning & Preparation

Column  
Formatting

Data  
Filtering/Sorting

Error Handling

Transform  
Columns

### Data Transformation

Reshape Data

Data  
Aggregation

Joining/Merging  
Queries

Custom  
Transformations



## Load

### Data Load

Load to Excel  
Workbook

Refresh Data

Power Pivot

Data Modeling

Enabling  
Features

Documentation & Applied Steps

Data Preview

Query Dependencies Viewer

M-Language and Functions

Advanced Editor

# Transpose

Reshape rows to columns and columns to rows.

| Item | Section | Patient1 | Patient2 |
|------|---------|----------|----------|
| 1    | A       | 0        | 10       |
| 2    | A       | 10       | 0        |
| 3    | A       | 10       | 1        |
| 4    | B       | 0        | 1        |
| 5    | B       | 1        | 0        |
| 6    | B       | 0        | 10       |

| Patient  | Item1 | Item2 | Item3 | Item4 | Item5 | Item6 |
|----------|-------|-------|-------|-------|-------|-------|
| Section  | A     | A     | A     | B     | B     | B     |
| Patient1 | 0     | 10    | 10    | 0     | 1     | 0     |
| Patient2 | 10    | 0     | 1     | 1     | 0     | 10    |



# Group By

- Aggregate data into groups based on specific column values (e.g., sum).



# Group By

- Aggregate data into groups based on specific column values (e.g., sum).
- Retains row and column orientation.

| Item | Section | Patient1 | Patient2 |
|------|---------|----------|----------|
| 1    | A       | 0        | 10       |
| 2    | A       | 10       | 0        |
| 3    | A       | 10       | 1        |
| 4    | B       | 0        | 1        |
| 5    | B       | 1        | 0        |
| 6    | B       | 0        | 10       |

| Section | Patient1 | Patient2 |
|---------|----------|----------|
| A       | 20       | 11       |
| B       | 1        | 11       |

# Pivot

- Reshape data and create a cross-tabulation by rotating rows into columns.

| Item | Section | Patient1 | Patient2 |
|------|---------|----------|----------|
| 1    | A       | 0        | 10       |
| 2    | A       | 10       | 0        |
| 3    | A       | 10       | 1        |
| 4    | B       | 0        | 1        |
| 5    | B       | 1        | 0        |
| 6    | B       | 0        | 10       |

| Patient   | Section A | Section B |
|-----------|-----------|-----------|
| Patient 1 | 0         | 0         |
| Patient 1 | 10        | 1         |
| Patient 1 | 10        | 0         |
| Patient 2 | 10        | 1         |
| Patient 2 | 0         | 0         |
| Patient 2 | 1         | 10        |

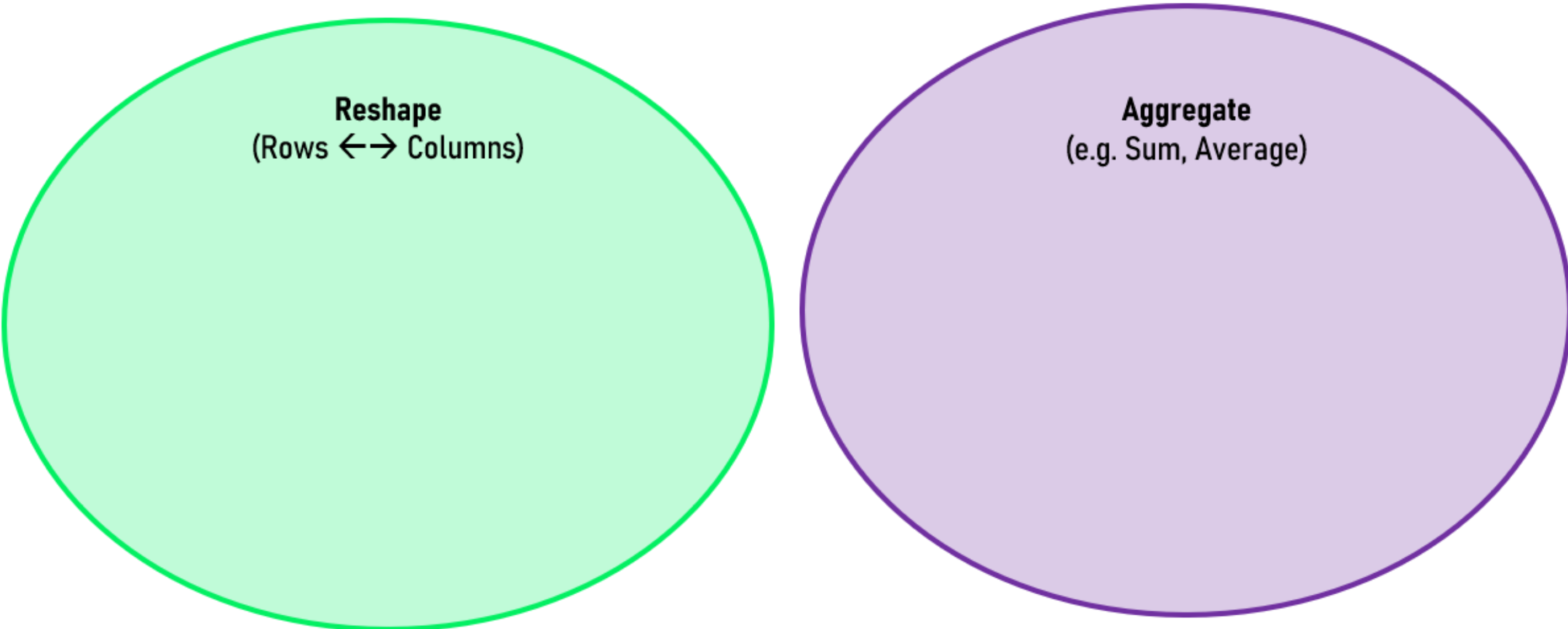
# Pivot

- Reshape data and create a cross-tabulation by rotating rows into columns.
- AND group/aggregate data based on specific columns (e.g., sum).

| Patient   | Section A | Section B |
|-----------|-----------|-----------|
| Patient 1 | 0         | 0         |
| Patient 1 | 10        | 1         |
| Patient 1 | 10        | 0         |
| Patient 2 | 10        | 1         |
| Patient 2 | 0         | 0         |
| Patient 2 | 1         | 10        |

| Patient   | Section A | Section B |
|-----------|-----------|-----------|
| Patient 1 | 20        | 1         |
| Patient 2 | 11        | 11        |

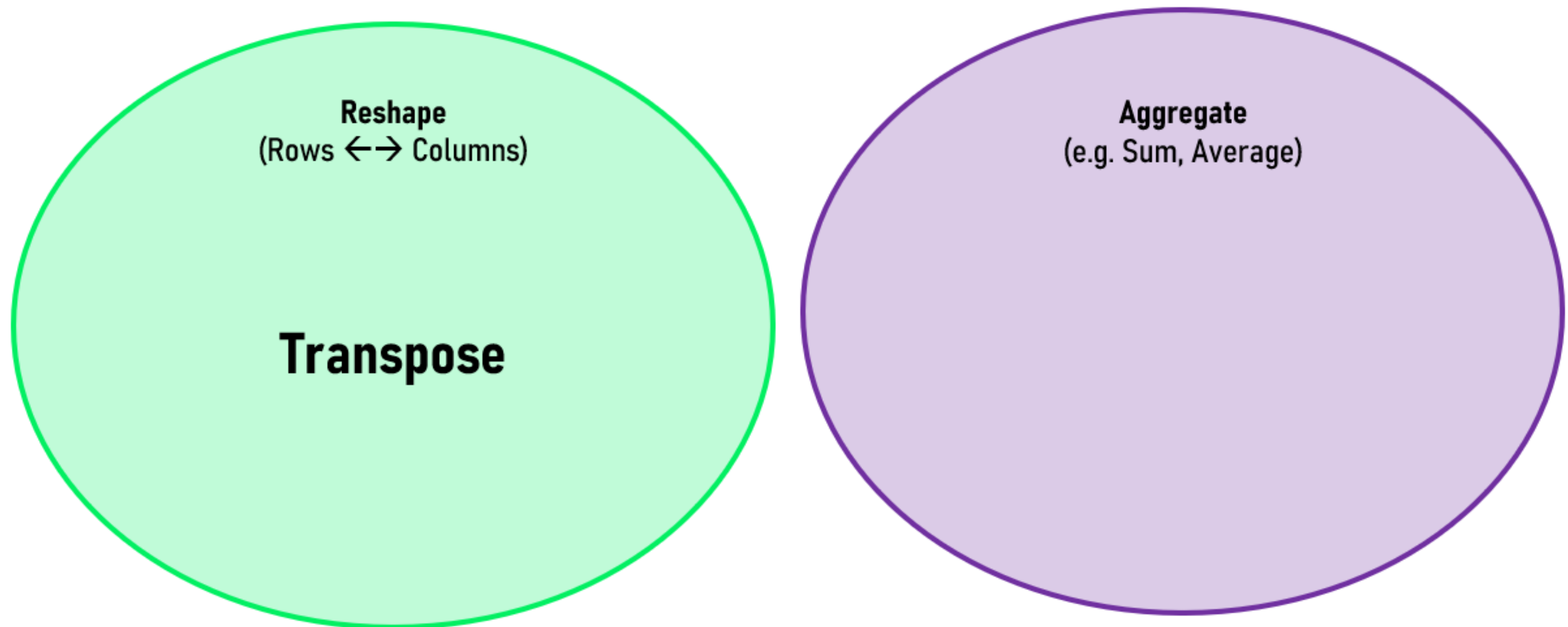
# Transpose vs. Pivot vs. Group By



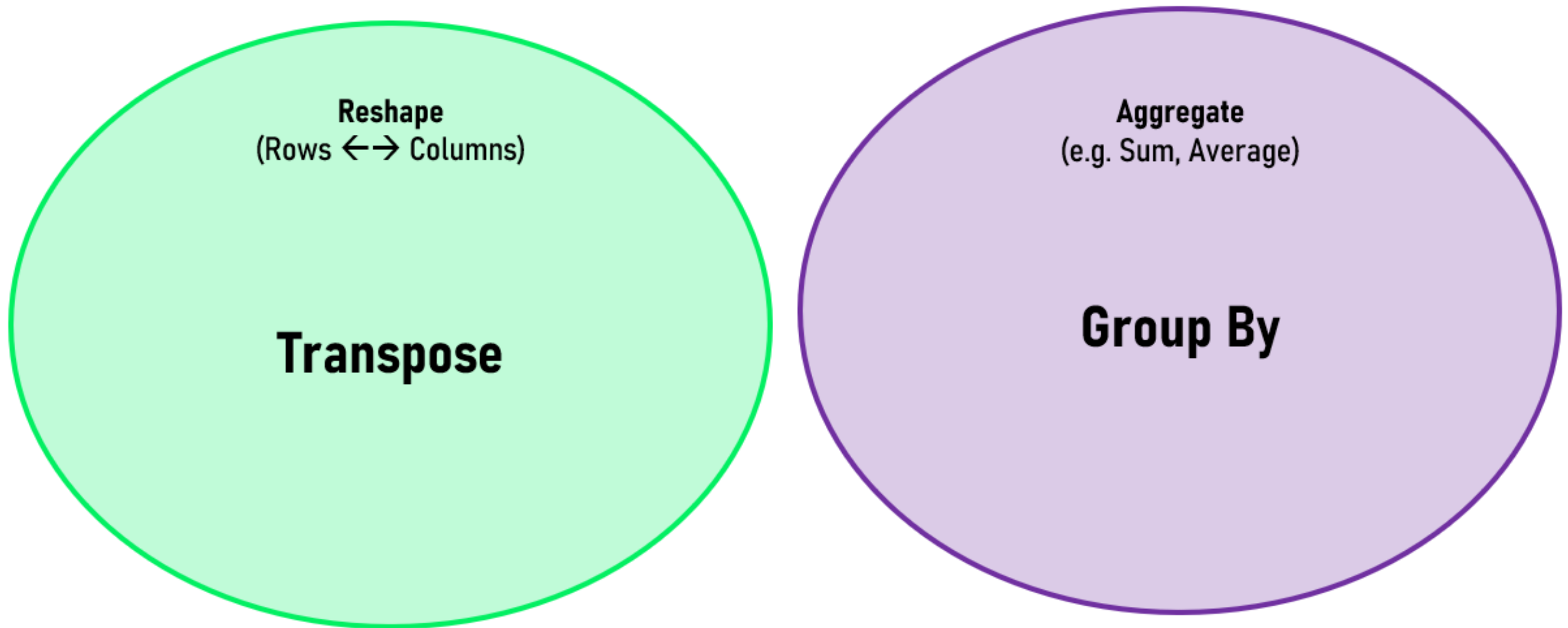
**Reshape**  
(Rows  $\leftrightarrow$  Columns)

**Aggregate**  
(e.g. Sum, Average)

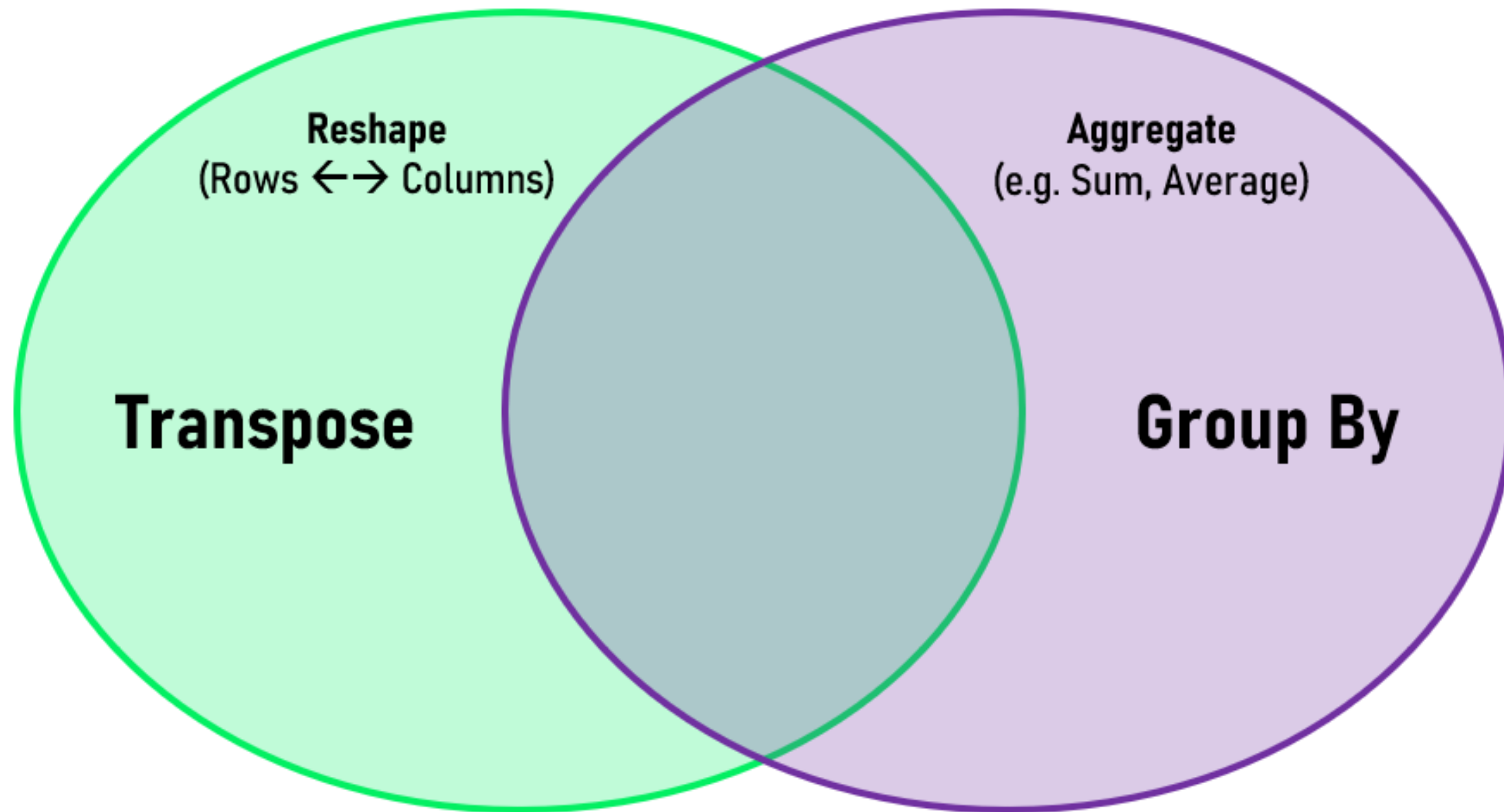
# Transpose vs. Pivot vs. Group By



# Transpose vs. Pivot vs. Group By

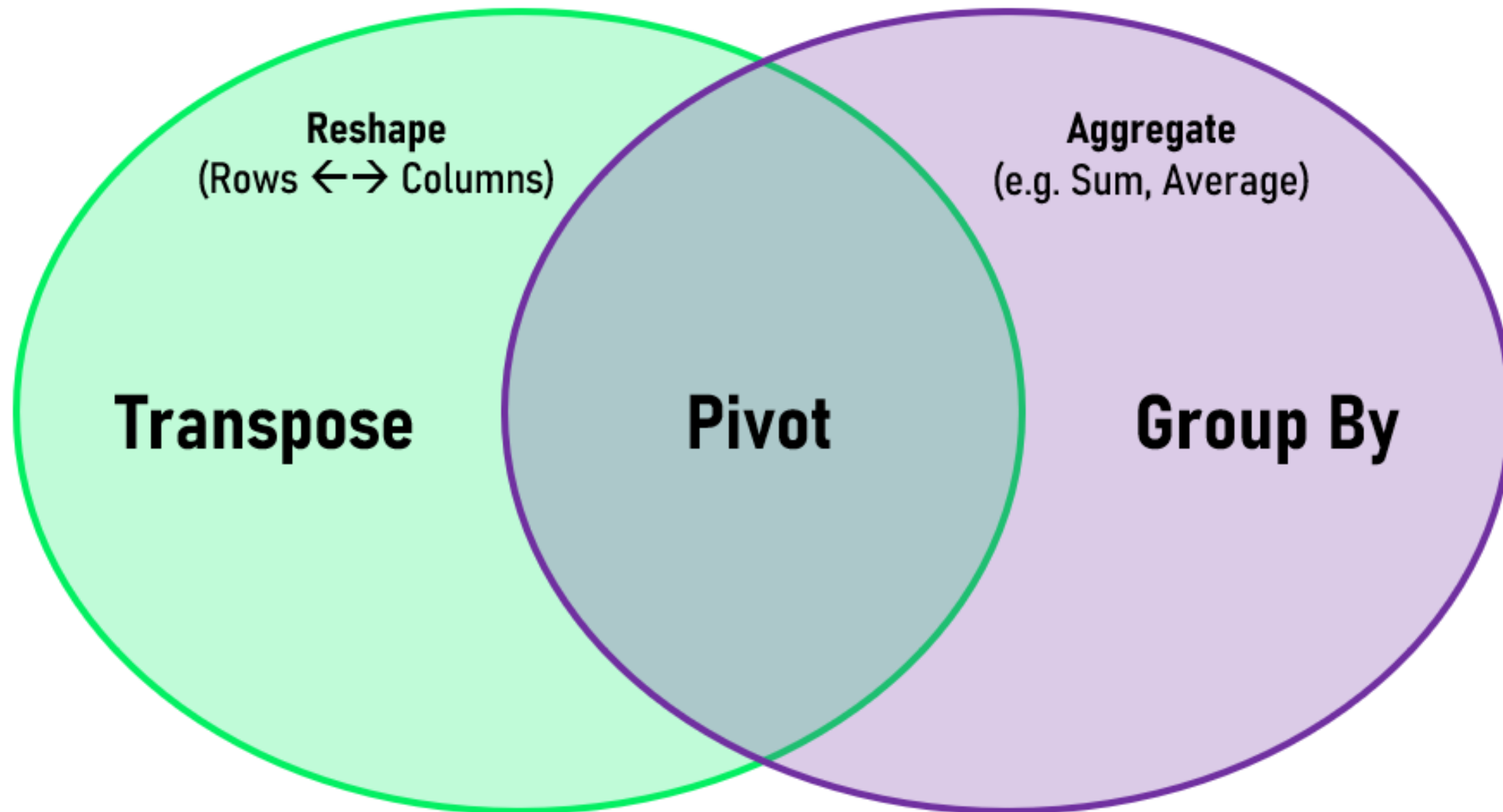


# Transpose vs. Pivot vs. Group By





# Transpose vs. Pivot vs. Group By





# Let's practice!

INTRODUCTION TO POWER QUERY IN EXCEL

# Shaping and Grouping Data in Excel Power Query

INTRODUCTION TO POWER QUERY IN EXCEL



**Full Name**

Performance Analytics Consultant

# Let's practice!

INTRODUCTION TO POWER QUERY IN EXCEL

# Congratulations!

INTRODUCTION TO POWER QUERY IN EXCEL

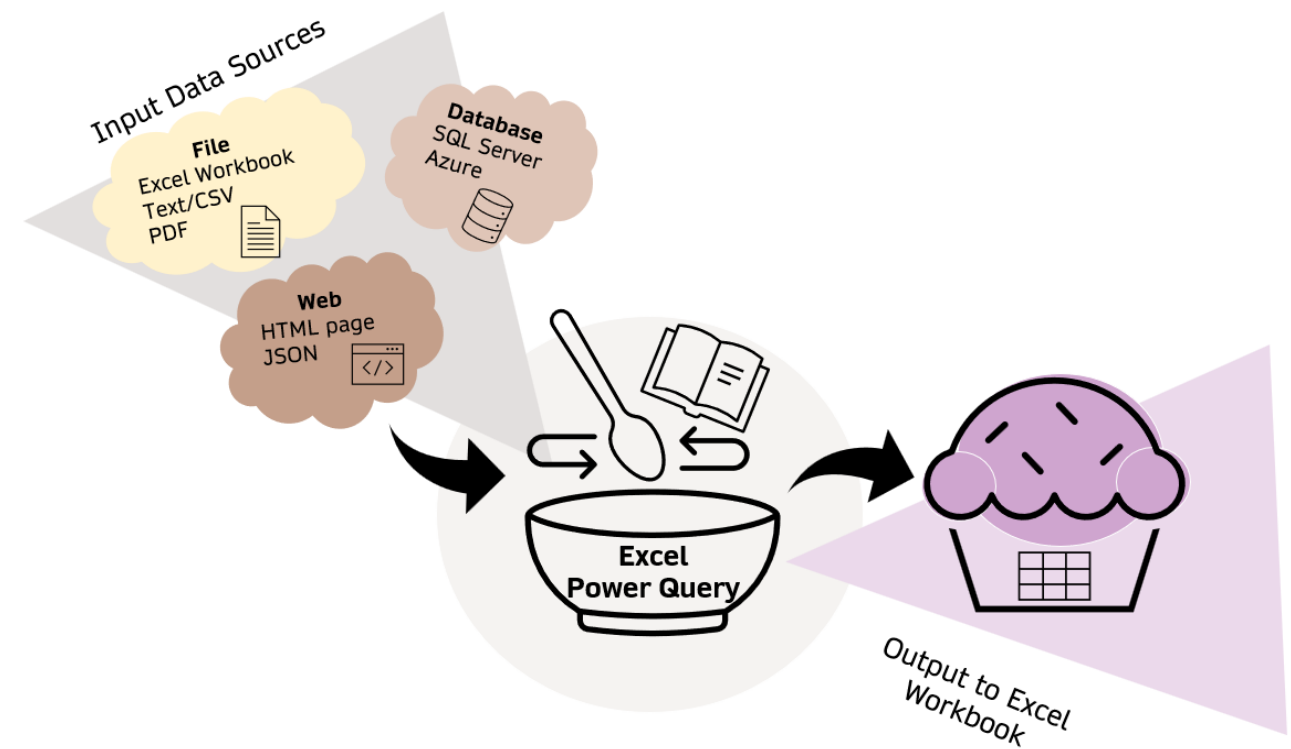


**Lyndsay Girard**

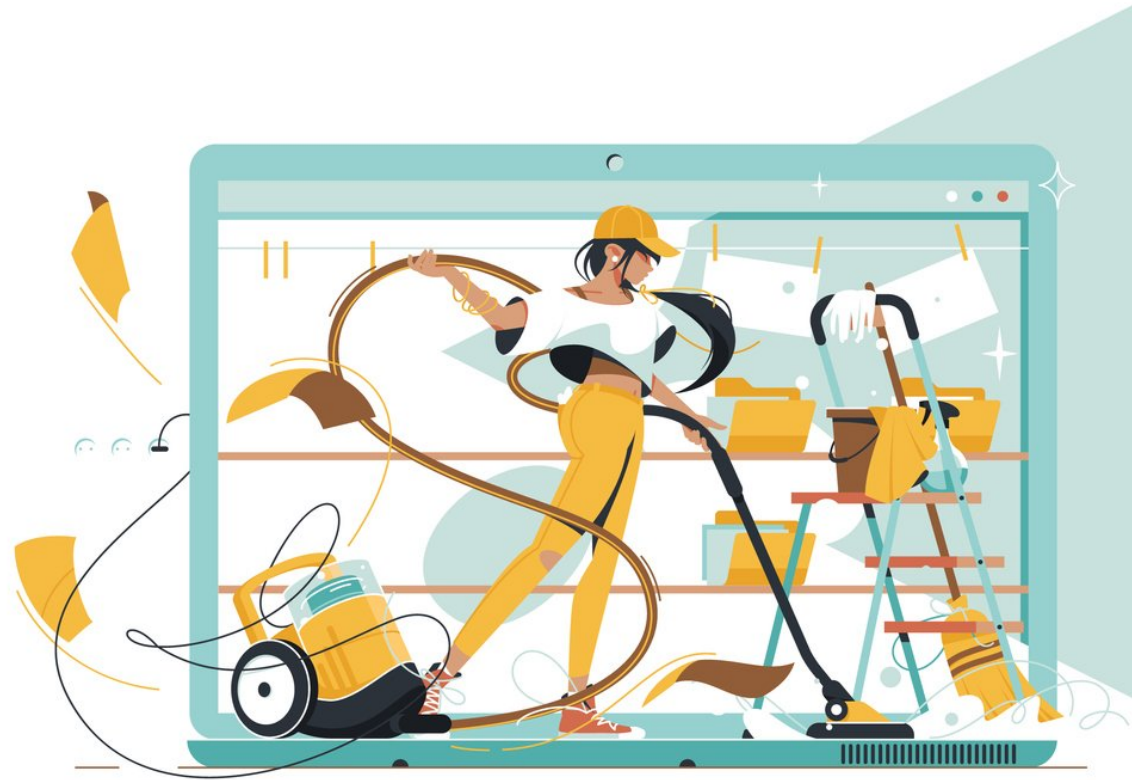
Performance Analytics Consultant

# Getting started with Power Query in Excel

- Main features of Power Query Editor.
- Power Query's role in ETL data workflows.
- Load data from multiple sources.
- Row and Column Manipulations.
- Applied Steps log.
- Loading tables into Excel Workbook.



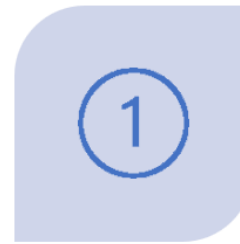
# Data Cleaning and Preparation



- Column profile/view column features.
- Data filtering.
- Troubleshoot errors and missing data.
- Removing duplicates.
- Handling outliers.

# Column Transformations

- Numerical transformations.
- Date/Time transformations.
- Text transformations.



NUMERIC



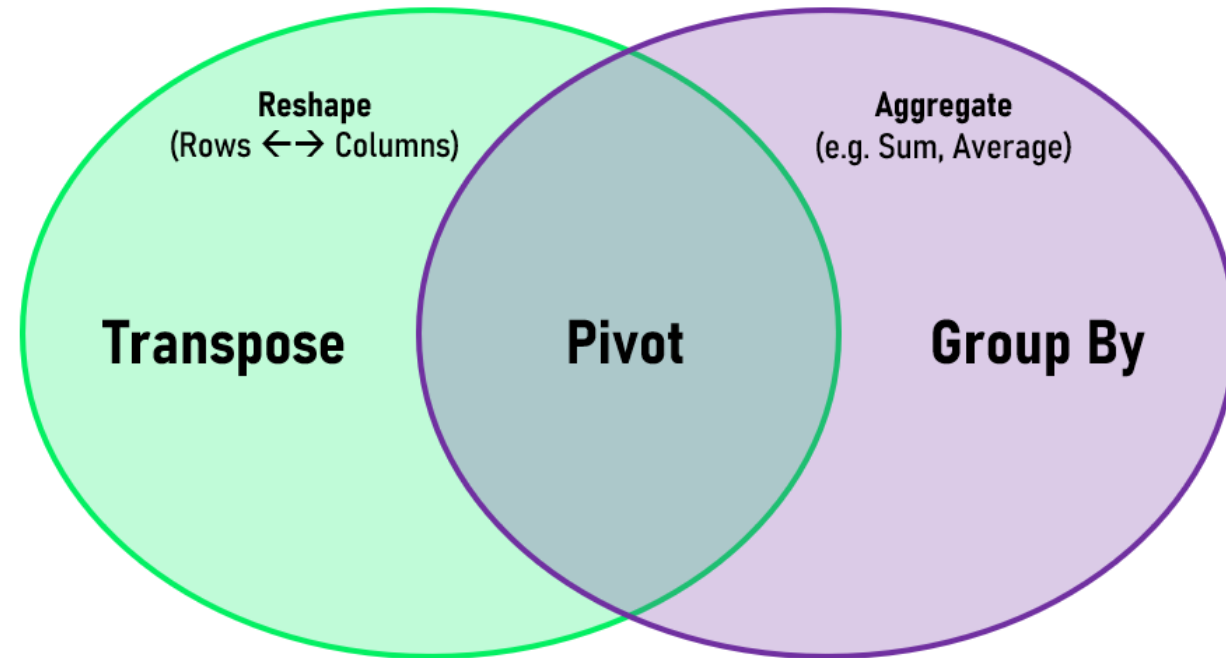
DATE/TIME



TEXT



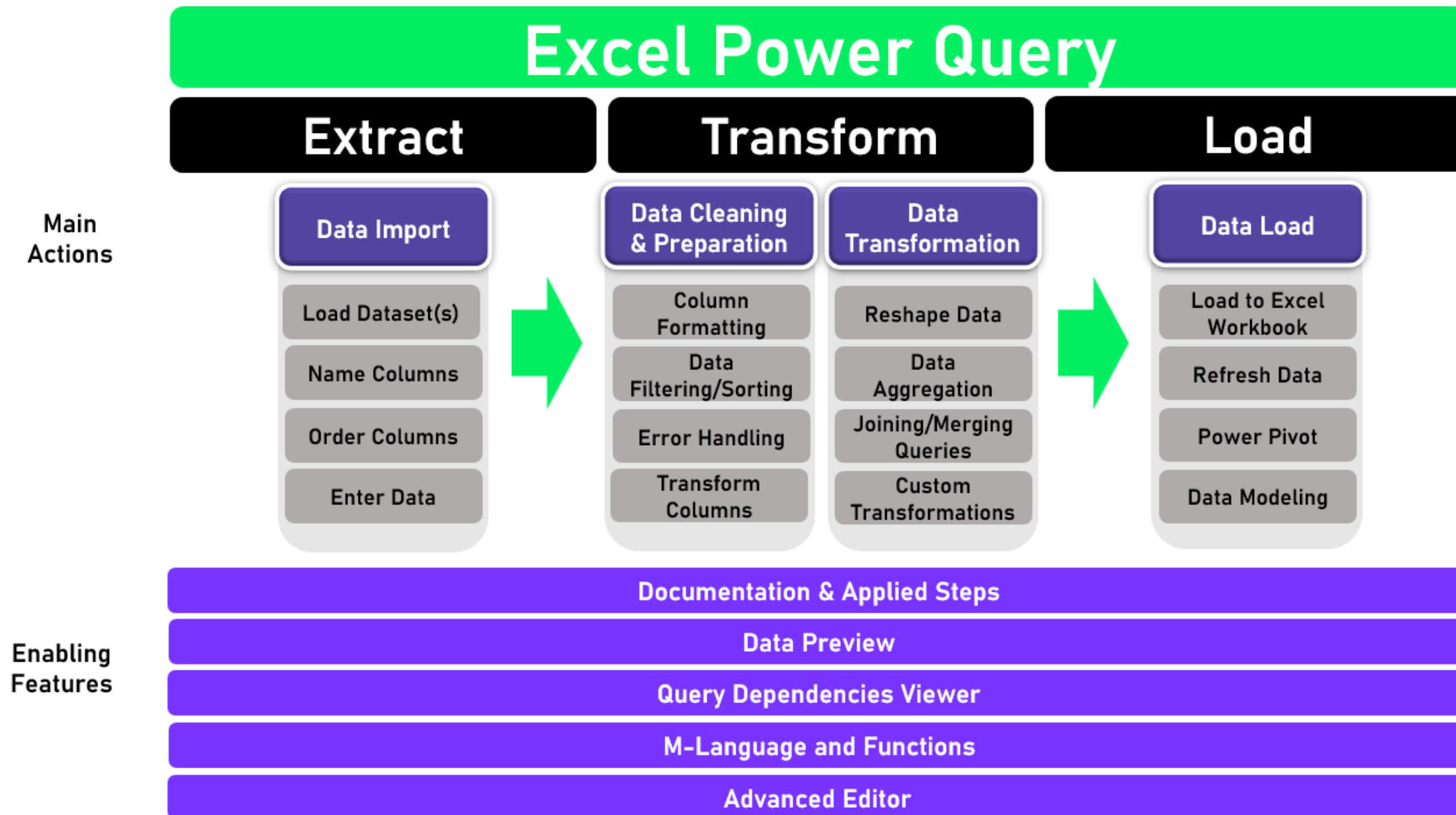
# Shaping and Grouping Data



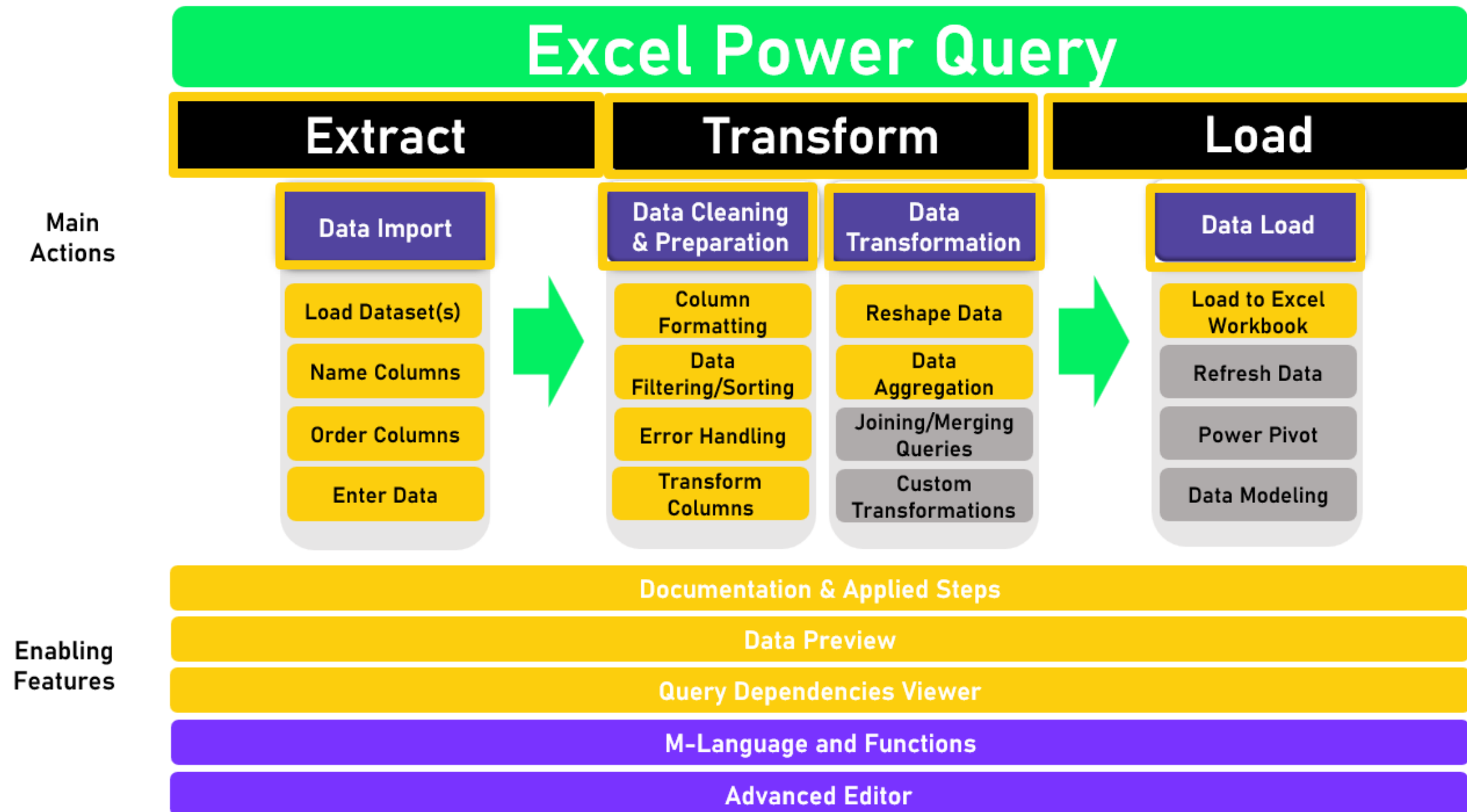
- Transpose transformations.
- Pivot transformations.
- Group By transformations.
- Reference vs. Duplicate query.



# Where we started



# What we have covered



# Congratulations!

INTRODUCTION TO POWER QUERY IN EXCEL